7+1: symbols and their definitions

The symbols of the notebook 7+1.nb and of its documentation, each with its defining equation

Repository: github.com/once-ere/7plus1 — prepared for Nash
Notebook: 7+1.nb (Wolfram Language 15.0.1)
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Abstract

This is a self-contained dictionary. For each symbol it gives the mathematical notation, the Mathematica name in 7+1.nb, the defining equation and its meaning, and, where it exists, the name used in the earlier notebook 4+4.nb. Sections 2–8 cover the octonions and the Clifford algebras (sections 0–6 of 7+1.nb); section 9 covers the Dirac equation and the Kohn–Sham theories of the field dirac16complex (sections 7–10 of 7+1.nb); section 10 covers gravity, the self-coupled field in gravitational fields and the Standard Model (sections 11–19 of 7+1.nb).

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1. Conventions used in every definition below

- Indices $A, B, K, L \in \{0, 1, \ldots, 7\}$ label the coordinates x_A, the octonion units e_A and the Clifford generators; $i, j \in \{1, \ldots, 7\}$; $h \in \{1, 2, 3\}$; $k \in \{4, \ldots, 7\}$ where stated.
- Mathematica lists are 1-based: the entry η_{AB} of a metric is `η[[A+1, B+1]]`.
- 1_n is the $n \times n$ identity matrix (`IdentityMatrix[n]`, named `ID4`, `ID8`, `ID16`, `ID32`); 0 in a block matrix is a zero block; M^{T} is the transpose, $\overline{M}$ the complex conjugate, $M^{\dagger}$ the conjugate transpose; δ_{AB} is the Kronecker delta; $\{X, Y\} = XY + YX$, $[X, Y] = XY - YX$.
- $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ is a block matrix (Mathematica `ArrayFlatten[{{a, b}, {c, d}}]`); $M \otimes N$ is the Kronecker product (`KroneckerProduct[M, N]`).
- An index 8 on a family of eight Clifford generators $G_0, \ldots, G_7$ means the ordered product $G_8 = G_0 G_1 \cdots G_7$ (the convention of 4+4.nb).
- $Cl(p, q)$: p generators square to $+1$ and q to -1 ($v^2 = Q(v)$, Q of signature (p, q)). This label is not the xAct list $\{7, 1, 0\}$, which declares seven space-like dimensions and one time-like dimension (section 2).

2. Coordinates and metrics

symbol	Mathematica	definition	meaning
X	`X`	$X = (x_0, x_1, \ldots, x_7)$	coordinates; also the components of a symbolic octonion or vector
x_0	`x0`	—	7+1.nb: ordinary time, octonion unit e_0. 4+4.nb: hidden space-like dimension
x_1, x_2, x_3	`x1, x2, x3`	—	ordinary 3-space (both notebooks); octonion units e_1, e_2, e_3
x_4	`x4`	—	7+1.nb: auxiliary space-like, $e_4 = \ell$. 4+4.nb: ordinary time
x_5, x_6, x_7	`x5, x6, x7`	—	7+1.nb: auxiliary space-like, e_5, e_6, e_7. 4+4.nb: auxiliary time-like ("superluminal deflating time")
$\eta^{(4,4)}$	`η4488`	$\mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$	metric of 4+4.nb (space-like +1, time-like −1); split-octonion norm
$\eta^{(7,1)}$	`η71`	$\mathrm{diag}(-1,1,1,1,1,1,1,1)$	7+1 metric in the sign convention of 4+4.nb (space-like +1): seven space-like and one time-like dimension, the xAct declaration $\{7,1,0\}$; generates $Cl(7,1)$
$\eta^{(1,7)}$	`η17`	$-\eta^{(7,1)} =$ $\mathrm{diag}(1,-1,-1,-1,-1,-1,-1,-1)$	simply the squares of the octonion basis: $\eta^{(1,7)}_{AB} = \mathrm{Re}(e_A e_B)$, $e_0^2 = +1$, $e_i^2 = -1$; $x^{\mathsf{T}}\eta^{(1,7)}x = \mathrm{Re}(xx)$; generates $Cl(1,7)$
$\{7,1,0\}$	`DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`	xAct signature list (section 12 of 7+1.nb, as in Einstein-Lovelock-7+1-Nash.nb)	seven space-like dimensions, one time-like dimension, none degenerate
$\eta^{AB}_{(1,7)}$	`η17inv`	$(\eta^{(1,7)})^{-1} = \eta^{(1,7)}$	inverse metric, raises indices
g^O	`gO`	1_8	norm metric of the octonions, $N(x) = x^{\mathsf{T}} g^O x$
g	`gAB44`	$\eta^{(4,4)}$	norm metric of the split octonions (4+4.nb: `g_AB`)
$\eta^{(5,3)}$	`η53`	$\mathrm{diag}(1,-1,-1,-1,1,1,1,1)$	$\mathrm{Re}(xx)$ of the split octonions

The signature: $\{7,1,0\}$ and $\mathrm{diag}(+1,-1,\ldots,-1)$. The 7+1 spacetime has one time-like dimension (x_0, the direction of the octonion unit $e_0 = 1$) and seven space-like dimensions (x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ auxiliary space). Two different objects describe this, and neither may be replaced by the other. (i) The tensor package xAct, used in section 12 of 7+1.nb exactly as in the author's notebook Einstein-Lovelock-7+1-Nash.nb, declares the metric with `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`: the list $\{7,1,0\}$ declares seven space-like dimensions, one time-like dimension and no degenerate dimension — the numbers of entries +1 and −1 of $\eta^{(7,1)} = \mathrm{diag}(-1,+1,\ldots,+1)$ (`η71`), the same spacetime written in the sign convention of 4+4.nb (space-like = +1, time-like = −1). (ii) The matrix $\eta^{(1,7)} = -\eta^{(7,1)} = \mathrm{diag}(+1,-1,\ldots,-1)$ (`η17`) is simply the squares of the octonion basis: $e_0^2 = +1$ for the unit $e_0 = 1$ (time) and $e_i^2 = -1$ for the seven imaginary units $e_1, \ldots, e_7$ (space); more generally it is the real part of the product of two basis units, and it gives the quadratic form of the octonions:

$$\eta^{(1,7)}_{AB} = \mathrm{Re}(e_A e_B), \qquad e_0^2 = +1, \qquad e_i^2 = -1 \quad (i = 1, \ldots, 7),$$
$$\mathrm{Re}(x\,x) = x_0^2 - x_1^2 - \cdots - x_7^2 = \eta^{(1,7)}_{AB}\, x^A x^B \qquad (x = x^A e_A \in \mathbb{O}).$$

$\eta^{(1,7)}$ gives the signs of the Clifford relation $\Gamma^{(1,7)}_A \Gamma^{(1,7)}_B + \Gamma^{(1,7)}_B \Gamma^{(1,7)}_A = 2\,\eta^{(1,7)}_{AB}\, 1_{16}$ of the real 16×16 Dirac matrices: time squares to $+1_{16}$ and each of the seven space directions to -1_{16}, exactly like the octonion units (in the notation $Cl(p,q)$, with p generators squaring to +1 and q to −1, the algebra $Cl(1,7) \cong \mathbb{R}(16)$). It also gives the signs of the metric components: $g_{00} > 0$ along the time-like direction, negative squares along the space-like directions. Every component computed by 7+1.nb follows from the component matrix of the metric; the declared list fixes for xAct the sign of the determinant of the metric, `SignDetOfMetric[g71]` $= -1$, which is

the same in both sign conventions, since $\det\eta^{(1,7)} = \det\eta^{(7,1)} = -1$. 7+1.nb checks `SignatureOfMetric[g71]` $= \{7,1,0\}$, `SignDetOfMetric[g71]` $= -1$, $\mathrm{Re}(e_A e_B) = \eta^{(1,7)}_{AB}$ and $\mathrm{Re}(x\,x) = \eta^{(1,7)}_{AB} x^A x^B$.

3. Objects reproduced from 4+4.nb

symbol	Mathematica	definition	meaning (4+4.nb name)
ϵ_{hpq4}	`Qa[h,p,q]`	`Signature[{h, p, q, 4}]`	sign of the permutation (Levi-Civita symbol)
	`Qb[h,p,q]`	$\delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4}$	auxiliary symbol
$(s_h)_{pq}$	`s4by4[h]`	$\epsilon_{hpq4} - (\delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4})$	self-dual antisymmetric 4×4 (`SelfDualAntiSymmetric`)
$(t_h)_{pq}$	`t4by4[h]`	$\epsilon_{hpq4} + (\delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4})$	anti-self-dual antisymmetric 4×4 (`AntiSelfDualAntiSymmetric`)
σ	`σ`	$\left(\begin{smallmatrix} 0 & 1_4 \\ 1_4 & 0 \end{smallmatrix}\right)$	split metric on spinor space (`σ`, `σ88`)
ζ_A	`ζ[A]`	$\zeta_0 = 1_8$; $\zeta_h = \left(\begin{smallmatrix} 0 & s_h \\ s_h & 0 \end{smallmatrix}\right)$; $\zeta_{7-h} = \left(\begin{smallmatrix} 0 & t_h \\ -t_h & 0 \end{smallmatrix}\right)$; $\zeta_7 = \zeta_1 \cdots \zeta_6$	8×8 building blocks (`ζ[A]`)
τ_A	`τ[A]`	$\tau_A = \zeta_A$	split-octonion matrices (`τ[A]`)
$\bar\tau_A$	`τbar[A]`	$\bar\tau_0 = 1_8$, $\bar\tau_A = \sigma\tau_A^{\mathsf T}\sigma$	conjugate matrices (symbolized tau-bar)
T_A	`T16[A]`	$\left(\begin{smallmatrix} 0 & \bar\tau_A \\ \tau_A & 0 \end{smallmatrix}\right)$, $A = 0,\ldots,7$	16×16 Dirac matrices, $\{T_A, T_B\} = 2\eta^{(4,4)}_{AB} 1_{16}$ (`T16^A[A]`)
T_8	`T16[8]`	$T_0 T_1 \cdots T_7 = \mathrm{diag}(-1_8, 1_8)$	chirality matrix (`T16^A[8]`)
σ_{16}	`σ16`	$T_0 T_1 T_2 T_3 = \mathrm{diag}(-\sigma, \sigma)$	(`σ16`)
P_L, P_R	`PL44`, `PR44`	$\frac{1}{2}(1_{16} - T_8)$, $\frac{1}{2}(1_{16} + T_8)$	chiral projectors (`P_L`, `P_R`)
U_h	`uBasis[[h]]`	eigenvectors of σ divided by $\sqrt{2}$ (order of `Eigensystem[σ]`)	orthonormal eigenbasis of σ (4+4.nb: double-struck u)
	`evalues`, `evecs`	`Eigensystem[σ]`	eigenvalues $(-1,-1,-1,-1,1,1,1,1)$ and eigenvectors of σ
u	`unit`	$U_8 = (1,0,0,0,1,0,0,0)^{\mathsf T}/\sqrt{2}$, `hUSE = 8`	unit spinor: $\sigma u = u$, $u^{\mathsf T} u = u^{\mathsf T}\sigma u = 1$ (`unit`, `hUSE`)
$E_{44}(h)$	`E44[h]`	$(E_{44})_{Ba} = (U_h^{\mathsf T}\sigma\bar\tau_B)_a$	vielbein (`E^A_a[h]`)
$E_{44}(h)^{-1}$	`E44inv[h]`	$(U_h^{\mathsf T}\sigma U_h)\,(\eta^{(4,4)}_{00}\tau_0 U_h, \ldots, \eta^{(4,4)}_{77}\tau_7 U_h)$ (columns)	inverse vielbein (`E^a_A[h]`)
ε_B	`eA[B]`, `EA`	$B = 1,\ldots,8$, `EA = Array[eA, 8]`	split-octonion units; $\varepsilon_B \leftrightarrow \tau_{B-1} \leftrightarrow x_{B-1}$
	`gridx44`	`Grid` of $\varepsilon_B\varepsilon_C = \sum_D (g_{BB} E_{44}\tau_{B-1} E_{44}^{-1})_{DC}\varepsilon_D$	split-octonion multiplication table (`gridx`)
	`gridx44From4p4nb`	the table printed by 4+4.nb, Out[325]	reference for the comparison check
L^S_A	`LS[A]`	$g_{AA}\, E_{44}(8)\,\tau_A\, E_{44}(8)^{-1}$	left multiplication by ε_{A+1}
$x \star_S y$	`sMul[x, y]`	$\left(\sum_A x_A L^S_A\right) y$	split-octonion product of coefficient lists
$\bar x$, $N_S(x)$	`sConj`, `sNorm2`	$(x_0, -x_1, \ldots, -x_7)$; $x^{\mathsf T}\eta^{(4,4)}x$	split conjugate and norm
	`zeroDivisor`	$(\varepsilon_1 + \varepsilon_5)(\varepsilon_1 - \varepsilon_5) = 0$	a product of two nonzero split octonions

4. The octonions

symbol	Mathematica	definition	meaning
τ^O_A	`τO[A]`	$\tau^O_0 = 1_8$; $\tau^O_A = \tau_A$ $(A = 1,2,3)$; $\tau^O_A = \tau_A\sigma$ $(A = 4,\ldots,7)$	octonion 8×8 matrices
$\bar\tau^O_A$	`τObar[A]`	$g^O(\tau^O_A)^{\mathsf T} g^O = (\tau^O_A)^{\mathsf T}$	conjugate matrices; $\tau^O_A\bar\tau^O_B + \tau^O_B\bar\tau^O_A = 2g^O_{AB}1_8$
E^O	`EO`	$(E^O)_{Ba} = (u^{\mathsf T} g^O \bar\tau^O_B)_a$; $E^O = \eta^{(4,4)} E_{44}(8)$	octonion vielbein (orthogonal)
$(E^O)^{-1}$	`EOinv`	$(E^O)^{\mathsf T} = (u^{\mathsf T} g^O u)(g^O_{00}\tau^O_0 u, \ldots, g^O_{77}\tau^O_7 u)$	inverse vielbein
e_A	`e[A]`, `eList`	$A = 0,\ldots,7$, `eList = Array[e, 8, 0]`	octonion units, $e_0 = 1$; $e_A \leftrightarrow \tau^O_A \leftrightarrow x_A$

symbol	Mathematica	definition	meaning
L_A^O	`LO[A]`	$g^O_{AA}\, E^O \tau^O_A (E^O)^{-1}$	left multiplication by e_A: $e_A e_K = \sum_L (L^O_A)_{LK} e_L$
	`gridO`	`Grid` of $e_A e_K$, same formula as `gridx`	octonion multiplication table
	`gridOExpected`	table computed by `tools/ref71.py`	reference for the comparison check
xy	`oMul[x, y]`	$\left(\sum_A x_A L^O_A\right) y$	octonion product of coefficient lists
$\bar{x}$	`oConj[x]`	$(x_0, -x_1, \ldots, -x_7)$	conjugate
$N(x)$	`oNorm2[x]`	$x^{\mathsf{T}} g^O x = x_0^2 + \cdots + x_7^2$	norm
$\mathrm{Re}\, x$	`oRe[x]`	x_0	real part
x^{-1}	`oInv[x]`	$\bar{x}/N(x)$	inverse
$[x, y, z]$	`oAssoc[x, y, z]`	$(xy)z - x(yz)$	associator
	`toE[x]`, `ev[A]`	$\sum_A x_A e_A$; `UnitVector[8, A+1]`	display; coefficient list of e_A
	`xs`, `ys`, `zs`	`Array[a, 8, 0]`, `Array[b, 8, 0]`, `Array[c, 8, 0]`	general symbolic octonions
	`fanoLines`, `orientedLines`	$\{i, j, k\}$ with $e_i e_j = \pm e_k$; (a, b, c) with $e_a e_b = +e_c$	the 7 Fano lines
	`fanoPicture`	centre e_1; vertices e_2, e_4, e_7; midpoints e_6, e_3, e_5	picture of the Fano plane
	`nonAssociativeTriples`	basis triples with $[e_A, e_B, e_K] \neq 0$	there are 168
	`xNum`, `yNum`, `zNum`	$e_0 + 2e_1 - e_4 + 3e_7$, $-2e_0 + e_2 + e_5$, $e_3 + e_6$	numerical example
ℓ	`ev[4]`	$\ell = e_4$	Cayley–Dickson doubling unit, $\ell^2 = -1$
pq, $\bar{p}$	`qMul`, `qConj`	product in $\mathrm{span}\{e_0, \ldots, e_3\}$; $(p_0, -p_1, -p_2, -p_3)$	quaternions ($e_1 e_2 = -e_3$)
(a, b)	`asPair[x]`	$x = a + \ell b$, $a = (x_0, \ldots, x_3)$, $b = (x_4, \ldots, x_7)$	octonion as a pair of quaternions
	`cdMul[γ]`	$(a, b)(c, d) = (ac + \gamma\, d\bar{b},\ \bar{a}d + cb)$	Cayley–Dickson; $\gamma = -1$ octonions, $+1$ split
	`cdPrompt`, `pm`	$(ac - \bar{d}b,\ \bar{a}d + cb)$	the formula of [1A] (fails)
	`promptX`, `promptY`	$e_0 + e_4 = (1, 1)$, $e_1 = (i, 0)$	its counterexample
s_{AB}	`sAB[A, B]`	-1 if $A \geq 4$ and $B \geq 4$, else $+1$	octonion product $= s_{AB} \times$ split product
c, φ	`cc`, `φ`	$c = (1, 1, 1, 1, i, i, i, i)$, $\varphi(x)_A = c_A x_A$	isomorphism split$\otimes\mathbb{C} \to \mathbb{O} \otimes \mathbb{C}$

5. Clifford algebras

symbol	Mathematica	definition	meaning
T_A^O	`T16O[A]`	$\begin{pmatrix} 0 & \bar{\tau}^O_A \\ \tau^O_A & 0 \end{pmatrix}$	$Cl(8,0)$: $\{T^O_A, T^O_B\} = 2\delta_{AB} 1_{16}$
T_8^O	`T16O[8]`	$T_0^O \cdots T_7^O = T_8 = \mathrm{diag}(-1_8, 1_8)$	volume element of $Cl(8,0)$
S_{44}	`S44`	$T_4 T_5 T_6 T_7 = \mathrm{diag}(\sigma, \sigma)$	$T^O_k = T_k S_{44}$ for $k \geq 4$
$\Gamma^{(1,7)}_A$	`Γ17[A]`	$\Gamma^{(1,7)}_0 = T^O_0 = T_0$; $\Gamma^{(1,7)}_i = T^O_0 T^O_i$ $(= T_0 T_i,\ i \leq 3$; $= T_0 T_i S_{44},\ i \geq 4)$	REAL 7+1 generators: $\{\Gamma^{(1,7)}_A, \Gamma^{(1,7)}_B\} = 2\eta^{(1,7)}_{AB} 1_{16}$
$\Gamma^{(1,7)}_8$	`Γ17[8]`	$\Gamma^{(1,7)}_0 \cdots \Gamma^{(1,7)}_7 = \begin{pmatrix} 0 & -1_8 \\ 1_8 & 0 \end{pmatrix}$	volume element, square -1_{16}
	`products256`	$\Gamma^{(1,7)}_{A_1} \cdots \Gamma^{(1,7)}_{A_k}$, $A_1 < \cdots < A_k$	the 256 basis products
τ^C_A, $\bar{\tau}^C_A$	`τC[A]`, `τCbar[A]`	$\tau^C_0 = \bar{\tau}^C_0 = i 1_8$; $\tau^C_k = \tau^O_k$, $\bar{\tau}^C_k = \bar{\tau}^O_k$	complexified 8×8 matrices
$\Gamma^{(7,1)}_A$	`Γ71[A]`	$\begin{pmatrix} 0 & \bar{\tau}^C_A \\ \tau^C_A & 0 \end{pmatrix}$; $\Gamma^{(7,1)}_0 = iT_0$, $\Gamma^{(7,1)}_k = T^O_k$	COMPLEX generators: $\{\Gamma^{(7,1)}_A, \Gamma^{(7,1)}_B\} = 2\eta^{(7,1)}_{AB} 1_{16}$
$\Gamma^{(7,1)}_8$	`Γ71[8]`	$\Gamma^{(7,1)}_0 \cdots \Gamma^{(7,1)}_7 = iT_8$	volume element, square -1_{16}
S	`Ssim`	$(1_{16} - i\Gamma^{(1,7)}_0)/\sqrt{2}$	unitary; $\Gamma^{(7,1)}_A = S(i\Gamma^{(1,7)}_A)S^{-1}$
	`intertwiners[gs]`	`NullSpace` basis of $\{M : M\overline{g} = gM\ \forall g\}$	Schur intertwiners
	`indicator[gs]`	sign of the scalar $M\overline{M}$	$+1$: $\mathbb{R}(16)$; -1: $\mathbb{H}(8)$
M_0	`M71`	intertwiners of $\Gamma^{(7,1)}$, spanned by $\Gamma^{(1,7)}_8$	$M_0\overline{M_0} = -1_{16}$
ε	`ε2`	$\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$	$\varepsilon^2 = -1_2$
$\widehat{\Gamma}^{(7,1)}_A$	`Γ71R32[A]`	$\varepsilon \otimes \Gamma^{(1,7)}_A$	REAL 32×32 generators with $\eta^{(7,1)}$
Q_i, Q_j, Q_k	`Qi`, `Qj`, `Qk`	$\varepsilon \otimes 1_{16}$, $\sigma_1 \otimes \Gamma^{(1,7)}_8$, $\sigma_3 \otimes \Gamma^{(1,7)}_8$	quaternion units commuting with all $\widehat{\Gamma}^{(7,1)}_A$

symbol	Mathematica	definition	meaning
	`commutantBasis`, `commutantTime`	`NullSpace` of $\{X : X\widehat{\Gamma}^{(7,1)}_A = \widehat{\Gamma}^{(7,1)}_A X\}$; its timing	dimension 4: $\mathbb{H}$
$\chi^{(7,1)}$	`χ71`	$-i\Gamma^{(7,1)}_8 = T_8$	chirality matrix ($\Gamma^{(7,1)}$ basis)
	`PL71`, `PR71`	$\frac{1}{2}(1_{16} \mp \chi^{(7,1)}) = P_L, P_R$	7+1 chiral projectors
$\chi^{(1,7)}$	`χ17`, `PL17`, `PR17`	$i\Gamma^{(1,7)}_8$; $\frac{1}{2}(1_{16} \mp \chi^{(1,7)})$	complex chirality and projectors (rank 8)
E^O_{16}	`EO16`	$\mathrm{diag}(E^O, E^O)$	rotation to the octonion basis
Γ^e_A	`Γ17e[A]`	$E^O_{16}\Gamma^{(1,7)}_A (E^O_{16})^{\mathsf{T}}$; $\Gamma^e_0 = \left(\begin{smallmatrix} 0 & 1_8 \\ 1_8 & 0 \end{smallmatrix}\right)$, $\Gamma^e_i = \mathrm{diag}(L^O_i, -L^O_i)$	$\Gamma^{(1,7)}$ in the octonion basis
$\vec{x}$, ψ	`xImag`, `ψs`	$(0, x_1, \ldots, x_7)$; `Array[ψ, 8, 0]`	imaginary part; a general octonion

6. Examples and bookkeeping

symbol	Mathematica	definition	meaning
Σ_{AB}	`Σ17[A, B]`, `pairs`	$\frac{1}{4}[\Gamma^{(1,7)}_A, \Gamma^{(1,7)}_B]$; `pairs` lists the 28 pairs $A < B$	generators of Spin(1,7)
$S(\beta)^{\pm 1}$	`Sβ`, `Sβinv`	$\cosh\frac{\beta}{2}\, 1_{16} \pm \sinh\frac{\beta}{2}\, \Gamma^{(1,7)}_0\Gamma^{(1,7)}_1$	boost along x_1 and its inverse
$R(\theta)^{\pm 1}$	`Rθ`, `Rθinv`	$\cos\frac{\theta}{2}\, 1_{16} \pm \sin\frac{\theta}{2}\, \Gamma^{(1,7)}_4\Gamma^{(1,7)}_5$	rotation in the x_4–x_5 plane and its inverse
Λ	`Λboost`, `Λrot`	$\Lambda_{BK} = \frac{1}{16}\mathrm{tr}(\eta^{(1,7)}_{BB}\Gamma^{(1,7)}_B S \Gamma^{(1,7)}_K S^{-1})$	Lorentz matrices of the two examples
Γ^A	`Γ17up[A]`	$\eta^{AA}_{(1,7)}\Gamma^{(1,7)}_A$	raised index
Ψ	`Ψfun`	$(f_1(X), \ldots, f_{16}(X))$, `f[k] @@ X`	16 arbitrary functions of $x_0, \ldots, x_7$
$\Gamma^A\partial_A$	`DiracOp17[ψ]`	$\sum_A \Gamma^A\, \partial\psi/\partial x_A$	7+1 Dirac operator
$\Box$	`waveOp17[ψ]`	$\sum_A \eta^{AA}_{(1,7)}\, \partial^2\psi/\partial x_A^2$	7+1 wave operator
p, m	`pvec`, `mass`	$p = (3,2,2,0,0,0,0,0)$ (contravariant p^A), $m = 1$	$p^{\mathsf{T}}\eta^{(1,7)}p = m^2$
$\not{p}$	`pslash`	$\sum_A p^A\Gamma^{(1,7)}_A$	$\not{p}^{\,2} = m^2 1_{16}$
W	`realAmplitudes`	`NullSpace[pslash - mass ID16]`	eight real spinor amplitudes, $(\not{p} - m)w = 0$
$p \cdot x$	`pvec.η17.X`	$p^{\mathsf{T}}\eta^{(1,7)}x$	phase of the plane waves
	`Ψwave`	$w\, e^{-i\, p\cdot x}$	solves $(i\Gamma^A\partial_A - m)\Psi = 0$; complex
J	`antiCommutant`	`NullSpace` of $J\Gamma^{(1,7)}_A + \Gamma^{(1,7)}_A J = 0$ $(A = 0, \ldots, 7)$	spanned by $\Gamma^{(1,7)}_8$ alone, $J^2 < 0$
k, $\not{k}$	`kvec`, `kslash`	$k = (1,1,0,0,0,0,0,0)$, $\not{k} = \sum_A k^A\Gamma^{(1,7)}_A$	null momentum, $\not{k}^{\,2} = 0$
	`realNull`	`NullSpace[kslash]`	8 real solutions of $\not{k}\, w = 0$
	`ΨmasslessReal`	$w\cos(k\cdot x)$	REAL solution of $\Gamma^A\partial_A\Psi = 0$
$\eta^{AB}_{(7,1)}$	`η71inv`	$(\eta^{(7,1)})^{-1} = \eta^{(7,1)}$	raises indices for $\widehat{\Gamma}$
$\widehat{\Gamma}^A\partial_A$	`DiracOp71R32[ψ]`	$\sum_A \eta^{AA}_{(7,1)}\widehat{\Gamma}^{(7,1)}_A\, \partial\psi/\partial x_A$	real 32-component Dirac operator
	`Ψfun32`	$(g_1(X), \ldots, g_{32}(X))$, `g32[k] @@ X`	32 arbitrary functions
P	`pslash32`	$\sum_A p^A\widehat{\Gamma}^{(7,1)}_A$	$P^2 = -m^2\, 1_{32}$
w	`w32`	`Array[w, 32]`	32 arbitrary real constants
	`Ψ32`	$w\cos(p\cdot x) + \frac{1}{m}Pw\sin(p\cdot x)$	REAL solution of $(\widehat{\Gamma}^A\partial_A - m)\Psi = 0$
$\Gamma^{(5,3)}_A$	`Γ53[A]`	$\Gamma^{(5,3)}_0 = T_0$, $\Gamma^{(5,3)}_i = T_0T_i$	real $Cl(5,3)$ with $\eta^{(5,3)}$
$\Gamma^{(0,8)}_A$	`Γ08[A]`	$T^O_8T^O_A$	real $Cl(0,8)$
	`gens[p]`	T^O_A for $A < p$, iT^O_A for $A \geq p$	complex generators of signature $(p, 8-p)$
	`classification`, `claimed1B`, `correct`	$(p, q,$ indicator$)$, $p = 8, \ldots, 0$; claimed and computed classes	classification table
	`checks`, `check[name, test]`	association of results; stores `TrueQ[test]` and returns `name -> result`	verification bookkeeping

symbol	Mathematica	definition	meaning
	`failedChecks`, `allChecksPassed`	names of checks that are not True; `AllTrue[Values[checks], TrueQ]`	must be `{}` and `True`

7. Explicit matrices

The 4×4 blocks, the seven split matrices τ_A, and the four new octonion matrices ($\tau_A^O = \tau_A$ for $A \le 3$):

$$s_1 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix},\ s_2 = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix},\ s_3 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix},$$

$$t_1 = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix},\ t_2 = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix},\ t_3 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix},$$

$$\tau_1 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix},\quad \tau_2 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\tau_3 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix},\quad \tau_4 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\tau_5 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix},\quad \tau_6 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\tau_7 = \begin{pmatrix} -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix},$$

$$\tau_4^O = \begin{pmatrix} 0&1&0&0&0&0&0&0\\ -1&0&0&0&0&0&0&0\\ 0&0&0&-1&0&0&0&0\\ 0&0&1&0&0&0&0&0\\ 0&0&0&0&0&-1&0&0\\ 0&0&0&0&1&0&0&0\\ 0&0&0&0&0&0&0&1\\ 0&0&0&0&0&0&-1&0 \end{pmatrix}, \quad \tau_5^O = \begin{pmatrix} 0&0&-1&0&0&0&0&0\\ 0&0&0&-1&0&0&0&0\\ 1&0&0&0&0&0&0&0\\ 0&1&0&0&0&0&0&0\\ 0&0&0&0&0&0&1&0\\ 0&0&0&0&0&0&0&1\\ 0&0&0&0&-1&0&0&0\\ 0&0&0&0&0&-1&0&0 \end{pmatrix},$$

$$\tau_6^O = \begin{pmatrix} 0&0&0&-1&0&0&0&0\\ 0&0&1&0&0&0&0&0\\ 0&-1&0&0&0&0&0&0\\ 1&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&1\\ 0&0&0&0&0&0&-1&0\\ 0&0&0&0&0&1&0&0\\ 0&0&0&0&-1&0&0&0 \end{pmatrix}, \quad \tau_7^O = \begin{pmatrix} 0&0&0&0&-1&0&0&0\\ 0&0&0&0&0&-1&0&0\\ 0&0&0&0&0&0&-1&0\\ 0&0&0&0&0&0&0&-1\\ 1&0&0&0&0&0&0&0\\ 0&1&0&0&0&0&0&0\\ 0&0&1&0&0&0&0&0\\ 0&0&0&1&0&0&0&0 \end{pmatrix}.$$

8. Old names (4+4.nb) and new names (7+1.nb)

4+4.nb	**7+1.nb**	**reason**
`σ`, `σ88`	`σ`	unchanged
`τ[A]`, `ζ[A]`, `s4by4[h]`, `t4by4[h]`	unchanged	—
symbolized tau-bar	`τbar[A]`	2-D symbol made with the Notation package; a plain name needs no package
`T16^A[A1]` (symbolized)	`T16[A]`	the superscript looked like a power and the letter A was not an index
`P_L`, `P_R`	`PL44`, `PR44`	the 7+1 projectors get their own names `PL71`, `PR71`, `PL17`, `PR17`
double-struck u	`uBasis`	plain name
`g_AB`	`gAB44`	subscripted name; plain name
`E^A_a[h]`, `E^a_A[h]` (symbolized)	`E44[h]`, `E44inv[h]`	the super- and subscripts were part of the name, not indices
`eA[1..8]` (1-based)	`e[0..7]` (0-based) for the octonions	aligns e_A with τ_A^O and x_A: no $B-1$ shift
`gridx`	`gridx44`, and the new `grid0`	two tables now

9. Sections 7–10 of 7+1.nb: the field dirac16complex and its Kohn–Sham theories

The symbols used for the Dirac equation and the Kohn–Sham theories of sections 7–10 of 7+1.nb. $\Gamma_A^{(1,7)}$, $\Gamma^A = \eta^{AA}_{(1,7)}\Gamma_A^{(1,7)}$, $\Gamma_8^{(1,7)}$ and $\chi^{(1,7)}$ are those of section 5 of this dictionary; here Γ_A means $\Gamma_A^{(1,7)}$, $\Gamma^0 = \Gamma_0^{(1,7)}$, and $\chi = \chi^{(1,7)}$.

symbol	**Mathematica**	**definition**	**meaning**
Ψ, Ψ_a	`dirac16complex`, `ψ16[a]`	$\Psi = (\Psi_1, \ldots, \Psi_{16})^{\mathsf{T}}$, `Table[ψ16[a] @@ X, {a, 1, 16}]`	quantized 16-component field; complex anticommuting operators
$\Psi_\uparrow$, $\Psi_\downarrow$	—	$\Psi = (\Psi_\uparrow; \Psi_\downarrow)$	upper and lower 8 components
$\bar\Psi$	—	$\Psi^\dagger\Gamma^0$	Dirac adjoint

symbol	Mathematica	definition	meaning
Ψ^c	—	$\Gamma_8^{(1,7)}\Psi^*$	charge conjugate; $(\Psi^c)^c=-\Psi$
$\delta^7(\vec{x}-\vec{y})$	—	$\int \delta^7(\vec{x}-\vec{y})f(\vec{y})\,d^7y=f(\vec{x})$	Dirac delta in the 7 space coordinates $\vec{x}=(x_1,\ldots,x_7)$
m	`mD`	$m>0$	mass; $m=1$ in the examples
$H_{\rm D}$	`hD[ψ, m]`	$-i\Gamma^0\Gamma^i\partial_i+m\Gamma^0$	Dirac Hamiltonian; Hermitian
$\vec{p}$, p^i, E	`pv7`, `Ep`	$\vec{p}=(p^1,\ldots,p^7)$, $E=\sqrt{m^2+\vec{p}^{\,2}}$	momentum and energy of a plane wave $e^{-i(Ex_0-\vec{p}\cdot\vec{x})}$
e_s	`UnitVector[8, s]`	$(e_s)_r=\delta_{rs}$	unit column of length 8
w_s	`uS[s, M, E]`	$\frac{1}{\sqrt{E}}\big(E\,e_s;\ (m1_8-p^i\tau_i^O)e_s\big)$	real positive-energy amplitudes, $(\not{p}-m)w_s=0$, $\not{p}=E\Gamma_0+p^i\Gamma_i$
$P_L^{(1,7)}$, $P_R^{(1,7)}$	`PL17`, `PR17`	$\frac{1}{2}(1_{16}\mp\chi^{(1,7)})$	complex chiral projections
Ψ_L, Ψ_R	`ϕL`, `ϕR`	$\frac{\Psi_\uparrow+i\Psi_\downarrow}{\sqrt{2}}$, $\frac{i\Psi_\uparrow+\Psi_\downarrow}{\sqrt{2}}$	chiral 8-component fields
g_S, g_P, g_V, g_A	`gS`, `gP`, `gV`, `gA`	real constants	scalar, pseudoscalar, vector, axial-vector contact couplings (mass dimension -6)
$\mathsf{s},\mathsf{p},\mathsf{v}_A,\mathsf{a}_A$	—	real functions of x	external sources
$\mathcal{L}$	—	below	Lagrangian density of the self-coupled field
ρ_S, ρ_P	`dens16[M]`	$\bar\Psi\Psi$, $\bar\Psi i\chi\Psi$	scalar and pseudoscalar densities
j^A, j_5^A	`dens16[M]`	$\bar\Psi\Gamma^A\Psi$, $\bar\Psi\Gamma^A\chi\Psi$; $j_A=\eta_{AA}^{(1,7)}j^A$	vector and axial-vector currents
Φ_k, λ_k	`orbV`	$H_{\rm KS}\Phi_k=\lambda_k\Phi_k$, $\int\Phi_k^\dagger\Phi_l=\delta_{kl}$	Kohn–Sham orbitals and eigenvalues
$H_{\rm KS}$	`hKS[p]`	below	Kohn–Sham Hamiltonian
$\mathcal{S},\mathcal{P},\mathcal{V}_A,\mathcal{A}_A$	`Sks`, `Pks`, `VKS`, `AKS`	below	Kohn–Sham potentials (lower index A)
F	—	$F[n]=\min_{\Psi\to n}\langle\Psi\vert H_0+H_{\rm int}\vert\Psi\rangle$	universal functional of the 18 densities $n=(\rho_S,\rho_P,j^A,j_5^A)$
T_s	—	$\sum_k\int\Phi_k^\dagger H_{\rm D}\Phi_k\,d^7x$	kinetic plus rest energy of the orbitals
E_H	`EH16`	below	Hartree energy
E_{xc}, E_x	`Exfun[R]`	$E_{xc}=F-T_s-E_H$; E_x below	exchange-correlation and exact exchange energies
$\mathcal{R}$	`Rorb`	$\sum_k\Phi_k\bar\Phi_k$	16×16 density matrix
Φ_{kL}, Φ_{kR}	`orbLV`, `orbRV`	$\frac{\Phi_{k\uparrow}+i\Phi_{k\downarrow}}{\sqrt{2}}$, $\frac{i\Phi_{k\uparrow}+\Phi_{k\downarrow}}{\sqrt{2}}$	chiral components of the orbitals
κ	`κc`	$\sum_k\Phi_{kL}^\dagger\Phi_{kR}$	complex scalar density: $\rho_S=\kappa+\kappa^*$, $\rho_P=i(\kappa-\kappa^*)$
$\mathcal{M}$	`Mks`	$m+\mathcal{S}+i\mathcal{P}$	complex Kohn–Sham mass
$\mathcal{V}_A^L$, $\mathcal{V}_A^R$	`VLks`, `VRks`	$\mathcal{V}_A-\mathcal{A}_A$, $\mathcal{V}_A+\mathcal{A}_A$	chiral vector potentials
ϑ	`Uθ`	$e^{i\vartheta\chi}=\cos\vartheta\,1_{16}-\sin\vartheta\,\Gamma_8^{(1,7)}$	chiral rotation angle; `Uθ` has $\cos\vartheta=\frac{3}{5}$
k_F, m^*	`kF0`, `gap7`	$m^*=m+g_S\rho_S(m^*)$	Fermi momentum and effective mass of uniform matter
ρ	`rho7`	$j^0=k_F^7/(105\pi^4)$	particle density of uniform matter
E/A, E_F	`eps7/rho7`, `fermi7`	$E_F=g_V\rho+\sqrt{k_F^2+m^{*2}}$	energy per particle and Fermi energy of uniform matter
$\Lambda_{\rm c}$	`Λc`	$\vert\vec{p}\vert<\Lambda_{\rm c}$, $\Lambda_{\rm c}=1$	momentum cutoff of the Dirac sea
g, g_c	`gNJL`, `gcrit`	$g=g_S=g_P$; $g_c=-90\pi^4/\Lambda_{\rm c}^6$	chirally symmetric coupling; critical coupling

Defining equations:

$$\big(i\Gamma^A\partial_A-m\big)\Psi=0,\qquad \big(i\partial_0+\tau_i^O\partial_i\big)\Psi_L=m\Psi_R,\qquad \big(i\partial_0-\tau_i^O\partial_i\big)\Psi_R=m\Psi_L,$$

$$\mathcal{L}=\bar\Psi\big(i\Gamma^A\partial_A-m\big)\Psi-\bar\Psi\big(\mathsf{s}+i\chi\mathsf{p}+\Gamma^A(\mathsf{v}_A+\chi\mathsf{a}_A)\big)\Psi$$
$$-\frac{g_S}{2}(\bar\Psi\Psi)^2-\frac{g_P}{2}(\bar\Psi i\chi\Psi)^2-\frac{g_V}{2}(\bar\Psi\Gamma^A\Psi)(\bar\Psi\Gamma_A\Psi)-\frac{g_A}{2}(\bar\Psi\Gamma^A\chi\Psi)(\bar\Psi\Gamma_A\chi\Psi),$$

$$H_{\rm KS}=\Gamma^0\Gamma^i\big(-i\partial_i+\mathcal{V}_i\big)+\Gamma^0(m+\mathcal{S})+\mathcal{V}_0+\Gamma^0 i\chi\mathcal{P}+\chi\mathcal{A}_0+\Gamma^0\Gamma^i\chi\mathcal{A}_i,$$

$$\mathcal{S}=\mathsf{s}+g_S\rho_S+\frac{\delta E_{xc}}{\delta\rho_S},\quad \mathcal{P}=\mathsf{p}+g_P\rho_P+\frac{\delta E_{xc}}{\delta\rho_P},\quad \mathcal{V}_A=\mathsf{v}_A+g_Vj_A+\frac{\delta E_{xc}}{\delta j^A},\quad \mathcal{A}_A=\mathsf{a}_A+g_Aj_{5A}+\frac{\delta E_{xc}}{\delta j_5^A},$$

$$E_H=\int\Big[\frac{g_S}{2}\rho_S^2+\frac{g_P}{2}\rho_P^2+\frac{g_V}{2}j^Aj_A+\frac{g_A}{2}j_5^Aj_{5A}\Big]d^7x,$$

$$E_x=-\frac{g_S}{2}\mathrm{tr}(\mathcal{R}\mathcal{R})-\frac{g_P}{2}\mathrm{tr}(i\chi\mathcal{R}\,i\chi\mathcal{R})-\frac{g_V}{2}\mathrm{tr}(\Gamma^A\mathcal{R}\Gamma_A\mathcal{R})-\frac{g_A}{2}\mathrm{tr}(\Gamma^A\chi\mathcal{R}\Gamma_A\chi\mathcal{R}),$$

$$\big[\mathcal{V}_0^L-\tau_i^O\partial_i-i\tau_i^O\mathcal{V}_i^L\big]\Phi_L+\mathcal{M}\Phi_R=\lambda\Phi_L,\qquad \mathcal{M}^*\Phi_L+\big[\mathcal{V}_0^R+\tau_i^O\partial_i+i\tau_i^O\mathcal{V}_i^R\big]\Phi_R=\lambda\Phi_R.$$

In $H_0 + H_{\rm int}$, $H_0 = \int :\Psi^\dagger H_{\rm D}\Psi: d^7x$ and $H_{\rm int} = \int \left[\frac{g_S}{2} :\rho_S^2: + \frac{g_P}{2} :\rho_P^2: + \frac{g_V}{2} :j^A j_A: + \frac{g_A}{2} :j_5^A j_{5A}:\right] d^7x$ (normal ordered); $\min_{\Psi\to n}$ runs over the states with densities n.

10. Sections 11–19 of 7+1.nb: gravity, the self-coupled field in gravitational fields, the Standard Model

The symbols of the third stage of the project (sections 11–19 of 7+1.nb). Here $\Gamma_A = \Gamma_A^{(1,7)}$, $\Gamma^A = \eta^{AA}_{(1,7)}\Gamma_A$, $\eta = \eta^{(1,7)}$, and Greek indices $\mu, \nu, \lambda, \rho, \sigma, \alpha, \beta, \gamma, \delta \in \{0, \ldots, 7\}$ are coordinate indices (on the brane $\mu, \nu \in \{0, \ldots, 3\}$).

symbol	Mathematica	definition	meaning
$g_{\mu\nu}$, $g^{\mu\nu}$	`cv["g"]`, `cv["ginv"]`	one time-like, seven space-like dimensions: xAct signature $\{7, 1, 0\}$; components with the signs of η ($g_{00} > 0$)	metric of 7+1 spacetime and its inverse
$\{^\lambda_{\mu\nu}\}$	`cv["chr"]`	(10.1)	Christoffel symbols
$R^\rho{}_{\sigma\mu\nu}$	`cv["riem"]`	(10.1)	Riemann tensor
$R_{\sigma\nu}$, R, $G^\mu{}_\nu$	`cv["ric"]`, `cv["R"]`, `cv["G"]`	(10.1)	Ricci tensor, scalar curvature, Einstein tensor
$R^{\gamma\delta}{}_{\alpha\beta}$	`cv["rud"]`	$g^{\delta\sigma}R^\gamma{}_{\sigma\alpha\beta}$	Riemann tensor, two indices up
$\delta^{\mu_1\cdots\mu_n}_{\nu_1\cdots\nu_n}$	`gKD`, `kδ`	$\det[\delta^{\mu_i}_{\nu_j}]$	generalized Kronecker delta
$E^{(k)\mu}{}_\nu$, L_k	`Lovelock1/2/3`, `lovelockE`, `lovelockL`	(10.2)	Lovelock tensors and densities
α_k, $\bar\Lambda$, κ, G_8	`alf[k]`, `lamb`, `kap`	(10.3)	couplings, cosmological constant, $\kappa = 8\pi G_8$
w_k, Λ, H	`w1`, `w2`, `w3`, `Λ`, `H`	(10.3)	pure-number couplings and inverse length of the reference notebook
$T^\mu{}_\nu$; ε, p_a, p_b	—; `ksStress`	$T =$ $\mathrm{diag}(\varepsilon, -p_a, -p_a, -p_a, -p_b, \ldots, -p_b)$	energy–momentum tensor; energy density and pressures
a, b, H_a, H_b	`a3[x0]`, `b4[x0]`, `Ha`, `Hb`	(10.4); $H_a = a'/a$, $H_b = b'/b$	scale factors and Hubble rates
q_H	`qH`	$b = e^{q_H H x_0}$ when $a = e^{H x_0}$	ratio of the Hubble rates of exponential solutions
p, q	`pK`, `qK`	$a = x_0^p$, $b = x_0^q$	Kasner exponents
$e^A{}_\mu$, $e_A{}^\mu$	`fd["e"]`, `fd["ei"]`	$g_{\mu\nu} = \eta_{AB}e^A{}_\mu e^B{}_\nu$	frame field and its inverse
$\Gamma_\mu(X)$, $\Gamma^\mu(X)$	`fd["Gam"]`, `fd["GamUp"]`	$e^A{}_\mu\Gamma_A$, $e_A{}^\mu\Gamma^A$	generalized frame matrices
$\omega_\mu{}^A{}_B$, Ω_μ	`fd["om"]`, `fd["Om"]`	(10.5)	canonical spin connection, spinor connection
D_μ, ∇_μ	—	$D_\mu = \partial_\mu + \Omega_\mu$; (10.5)	spinor covariant derivative; total covariant derivative
$R_{AB\mu\nu}$	`rABmn`	$\eta_{AC}e^C{}_\rho R^\rho{}_{\sigma\mu\nu}e_B{}^\sigma$	Riemann tensor with frame indices
q, A_μ	`qG`, `aGauge`	$D_\mu = \partial_\mu + \Omega_\mu - iqA_\mu$	charge and abelian gauge field
M (pair creation), $\lvert\beta\rvert^2$	`bogoliubov`	$\lvert\beta\rvert^2 = 1/(e^{2\pi M/H}+1)$ in de Sitter space	mass of a mode; pair-creation probability
K, Θ^K, Θ_K	`mIdx`, `θup[K]`, `θdn[K]`	(10.6)	multi-index and the 256 Clifford products
g_r; g_S, g_V, g_A, g_P	`gg[r]`; `gS`, `gV`, `gA`, `gP`	$g_0 = g_S$, $g_1 = g_V$, $g_7 = -g_A$, $g_8 = -g_P$	self-couplings of rank r; the four of section 11.3
F_{rs}, G_s	`fierzF`, `Geff`	(10.6)	Fierz matrix and Hartree–Fock couplings
M (Kohn–Sham), ρ_S, ρ	`ksGap`, `ksMoments`	$M = m + G_0\rho_S$	effective mass, scalar density, density of uniform matter
W, $h_{\mu\nu}$, L_c, b_0	—	$W = \{x_4 = \cdots = x_7 = 0\}$, $h_{\mu\nu} = g_{\mu\nu}(x, y = 0)$	world brane, induced metric, torus period, static scale
S, $S_{\rm EL}$, S_Ψ, $S_{\rm stab}$	—	(10.9)	total action; Einstein–Lovelock, dirac16complex and stabilizing parts
$\mathcal{L}_{\rm self}$, $\mathcal{L}_{\rm SM}$, $\mathcal{L}_{\rm portal}$	`LselfS`	(10.9); $\mathcal{L}_{\rm self} = -\frac{1}{2}\sum_r g_r I_r$	self-coupling, Standard Model and portal Lagrangians
$V_\perp$, G_4	`vPerp`	$V_\perp = b_0^4 L_c^4$, $G_4 = G_8/V_\perp$	proper volume of the torus; four-dimensional Newton constant
v, G_F, J	`vEW`, `gFermi`, `jCKM`	$v = (\sqrt{2}G_F)^{-1/2}$; $J = \mathrm{Im}(V_{us}V_{cb}V_{ub}^*V_{cs}^*)$	Higgs vacuum value, Fermi constant, Jarlskog invariant

symbol	Mathematica	definition	meaning
$V_{\rm CKM}$, $U_{\rm PMNS}$, U	`vCKM`, `uPMNS`	standard parametrization	quark and lepton mixing matrices
ρ_4, Λ_4	—	$3H_a^2 = 8\pi G_4\rho_4 + \Lambda_4$	energy density and cosmological constant of the brane
γ_μ, γ_5, ρ_k	`g4D`, `g5D`, `rhoK`	(10.7)	brane Dirac matrices and transverse matrices
R_A, U (brane)	`rBr`, `uInt`	(10.7)	brane form of the Dirac matrices; intertwiner
ψ_a, e_a	—	$\Psi = U\sum_a \psi_a \otimes e_a$	four brane Dirac spinors; unit vectors of $\mathbb{C}^4$
$\vec{q}$	`qKK`	$\vec{q} = 2\pi\vec{n}/L_c$	auxiliary momentum
μ, m, N	`muPortal`, `mPsi`, `nAct`	$1 - NN^\dagger = \mu\mu^\dagger/m^2$	portal Dirac mass, dirac16complex mass, light mixing matrix
$P(\alpha\to\beta)$, X_{ij}, Δm^2_{ij}	`pVac`, `xFormula`, `dm21`, `dm31`	(10.8)	oscillation probability
L, E, n_e	—		baseline, neutrino energy, electron density

$$\{^{\lambda}_{\mu\nu}\} = \tfrac{1}{2}g^{\lambda\rho}(\partial_\mu g_{\rho\nu} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}), \tag{10.1}$$

$$R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu\{^{\rho}_{\sigma\nu}\} - \partial_\nu\{^{\rho}_{\sigma\mu}\} + \{^{\rho}_{\mu\lambda}\}\{^{\lambda}_{\sigma\nu}\} - \{^{\rho}_{\nu\lambda}\}\{^{\lambda}_{\sigma\mu}\},$$

$$R_{\sigma\nu} = R^{\rho}{}_{\sigma\rho\nu}, \quad G^{\mu}{}_{\nu} = R^{\mu}{}_{\nu} - \tfrac{1}{2}\delta^{\mu}_{\nu}R,$$

$$E^{(k)\mu}{}_{\nu} = -\frac{1}{2^{k+1}}\delta^{\mu\alpha_1\beta_1\cdots\alpha_k\beta_k}_{\nu\gamma_1\delta_1\cdots\gamma_k\delta_k} R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1}\cdots R^{\gamma_k\delta_k}{}_{\alpha_k\beta_k}, \tag{10.2}$$

$$L_k = \frac{1}{2^k}\delta^{\alpha_1\beta_1\cdots\alpha_k\beta_k}_{\gamma_1\delta_1\cdots\gamma_k\delta_k} R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1}\cdots R^{\gamma_k\delta_k}{}_{\alpha_k\beta_k},$$

$$0 = -\Lambda\delta^{\mu}_{\nu} + \sum_{k=1}^{3} H^{-2k} w_k E^{(k)\mu}{}_{\nu}, \quad \sum_{k=0}^{3}\alpha_k E^{(k)\mu}{}_{\nu} = \kappa T^{\mu}{}_{\nu}, \tag{10.3}$$

$$\alpha_0 = 2\bar{\Lambda},\ \alpha_1 = 1,\ \alpha_2 = \tfrac{w_2}{w_1 H^2},\ \alpha_3 = \tfrac{w_3}{w_1 H^4},$$

$$ds^2 = dx_0^2 - a^2(dx_1^2 + dx_2^2 + dx_3^2) - b^2(dx_4^2 + \cdots + dx_7^2), \tag{10.4}$$

$$\omega_\mu{}^{A}{}_{B} = e^{A}{}_{\nu}\big(\partial_\mu e_B{}^{\nu} + \{^{\nu}_{\mu\lambda}\}e_B{}^{\lambda}\big), \quad \Omega_\mu = \tfrac{1}{4}\eta_{AC}\omega_\mu{}^{C}{}_{B}\Gamma^A\Gamma^B, \tag{10.5}$$

$$\nabla_\mu\Gamma_\nu = \partial_\mu\Gamma_\nu - \{^{\lambda}_{\mu\nu}\}\Gamma_\lambda + [\Omega_\mu, \Gamma_\nu] = 0,$$

$$\Theta^K = i^{\sigma_r}\Gamma^{A_1}\cdots\Gamma^{A_r}, \quad \sigma_r = \tfrac{r(r-1)}{2} \bmod 2, \quad \Theta_K = \eta_{A_1A_1}\cdots\eta_{A_rA_r}\Theta^K, \tag{10.6}$$

$$F_{rs} = \tfrac{1}{16}\sum_l (-1)^{rs-l}\tbinom{s}{l}\tbinom{8-s}{r-l}, \quad G = (1 - F^{\mathsf{T}})g,$$

$$\gamma_0 = \mathrm{diag}(1,1,-1,-1),\ \gamma_i = \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix},\ \rho_4 = i\gamma_0,\ \rho_{4+i} = -\gamma_i, \tag{10.7}$$

$$R_\mu = \gamma_\mu \otimes 1_4,\ R_k = \gamma_5 \otimes \rho_k,\ \Gamma_A U = U R_A,$$

$$P(\alpha\to\beta) = \Big|\sum_i U_{\beta i}U^*_{\alpha i}e^{-im_i^2 L/(2E)}\Big|^2, \quad X_{ij} = U^*_{\alpha i}U_{\beta i}U_{\alpha j}U^*_{\beta j}, \quad \Delta m^2_{ij} = m_i^2 - m_j^2, \tag{10.8}$$

$$S = S_{\rm EL} + S_\Psi + \int_W d^4x\sqrt{|h|}(\mathcal{L}_{\rm SM} + \mathcal{L}_{\rm portal}) + S_{\rm stab}, \tag{10.9}$$

$$S_{\rm EL} = -\tfrac{1}{2\kappa}\int d^8X\sqrt{|g|}\sum_k \alpha_k L_k, \quad S_\Psi = \int d^8X\sqrt{|g|}\big[\bar{\Psi}(i\Gamma^\mu D_\mu - m)\Psi + \mathcal{L}_{\rm self}\big].$$